

		$\Rightarrow v = 5.6 \text{ m s}^{-1}$
4	B	Conservation of Linear Momentum: $0.113u_1 + 0 = 0.113v_1 + 1(3.8 \times 10^6)$ Elastic: $u_1 - 0 = 3.8 \times 10^6 - v_1$ Thus, $u_1 = 1.87 \times 10^7 \text{ m s}^{-1}$
5	D	From the diagram, it is obvious that the net force is not zero (all forces